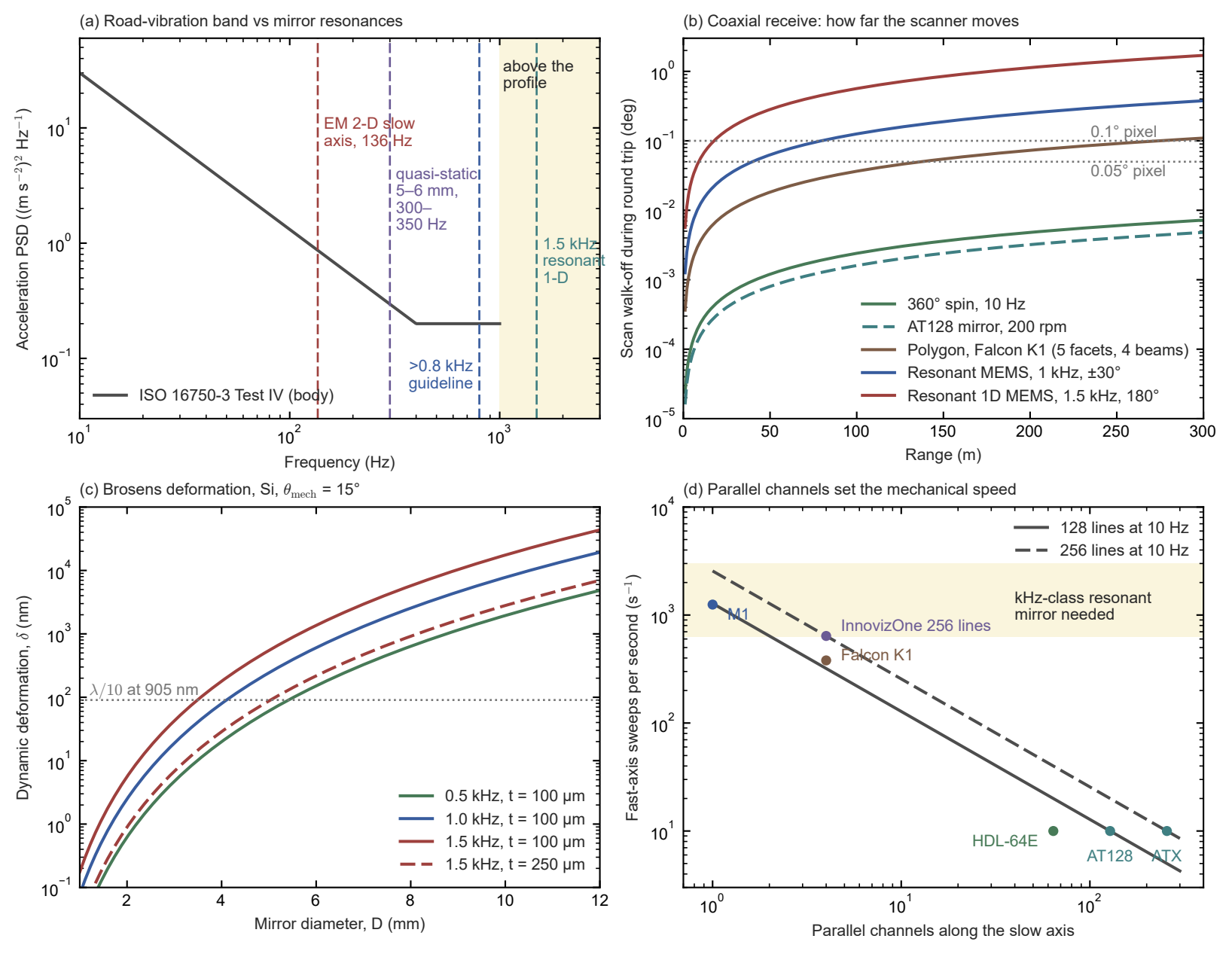

(a) Road-vibration band vs mirror resonances
Acceleration PSD ((m s^{-2})^2 Hz^{-1})
Frequency (Hz)
ISO 16750-3 Test IV (body)
EM 2-D slow axis, 136 Hz
quasi-static 5–6 mm, 300–350 Hz
>0.8 kHz guideline
1.5 kHz resonant 1-D
above the profile
(b) Coaxial receive: how far the scanner moves
Scan walk-off during round trip (deg)
Range (m)
0.1° pixel
0.05° pixel
360° spin, 10 Hz
AT128 mirror, 200 rpm
Polygon, Falcon K1 (5 facets, 4 beams)
Resonant MEMS, 1 kHz, ±30°
Resonant 1D MEMS, 1.5 kHz, 180°
(c) Brosens deformation, Si, θ_{mech} = 15°
Dynamic deformation, δ (nm)
Mirror diameter, D (mm)
$\lambda/10$ at 905 nm
0.5 kHz, t = 100 µm
1.0 kHz, t = 100 µm
1.5 kHz, t = 100 µm
1.5 kHz, t = 250 µm
(d) Parallel channels set the mechanical speed
Fast-axis sweeps per second (s^{-1})
Parallel channels along the slow axis
128 lines at 10 Hz
256 lines at 10 Hz
kHz-class resonant mirror needed
M1
InnovizOne 256 lines
Falcon K1
HDL-64E
AT128
ATX